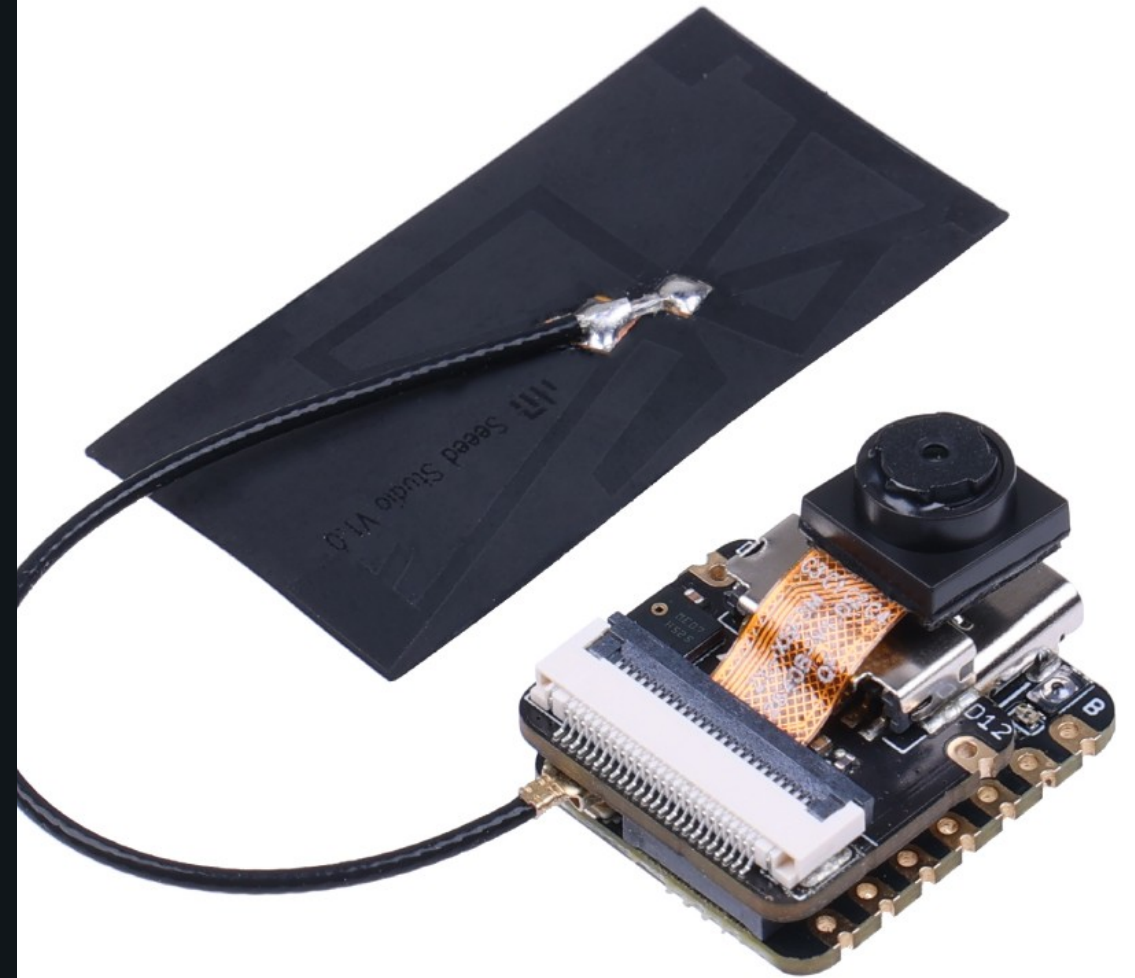


ECE 116

Advanced Hardware & Software Design

Intelligent embedded systems | Fall 2026



Instructional Team

Course staff, office hours, room, and meeting time are published on Canvas.



Professor Edward Wang

ejaywang@ucsd.edu



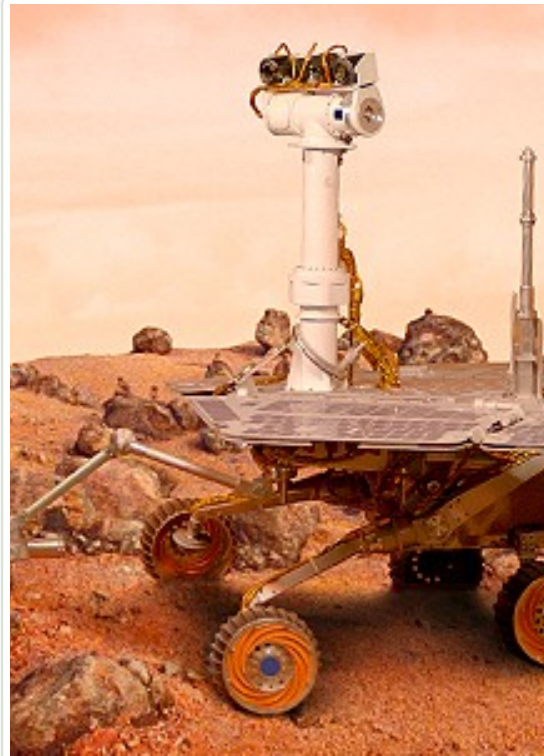
Pamely Zantou

pzantou@ucsd.edu

Intelligent systems live where software meets the physical world



health



exploration



assistive care



automation

Different applications. The same need to connect computation to real conditions and consequences.

A useful device must observe, decide, and act

observe a request
or condition

maintain state
and make a decision



act in the
physical world

make the result
observable

The difficult bugs often live in the interfaces between these responsibilities.

The XIAO is our common hardware platform



- camera
- microphone
- GPIO, ADC, and buses
- USB power, upload, and serial

One board lets us change the sensing, decision, and communication capability without changing platforms.

Assignments

GL5528 LDR voltage divider

Brighter light lowers the LDR resistance, so the A0 node rises with light and falls when covered. Confirm the direction and dark level on YOUR board.

A1

Microcontroller &
Finite State Machine

Time (seconds)

A2

Running and
Evaluating a Model

A3

Training a Vision
Model

XIAO ESP32-S3 pins used in A1

D4/D5 are the default I2C pins on the XIAO ESP32-S3. Pin labels (D1..D5) are the silk names; the sketch uses the underlying GPIO numbers shown beside each.

A4

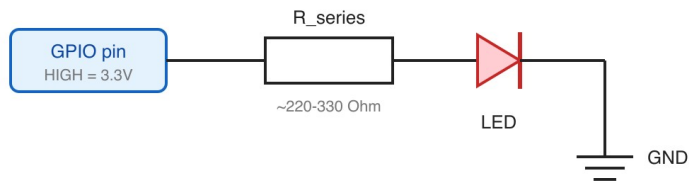
Edge Deployment

A5

Cloud Connectivity

Assignments

Mode LED with current-limiting resistor (one per LED)

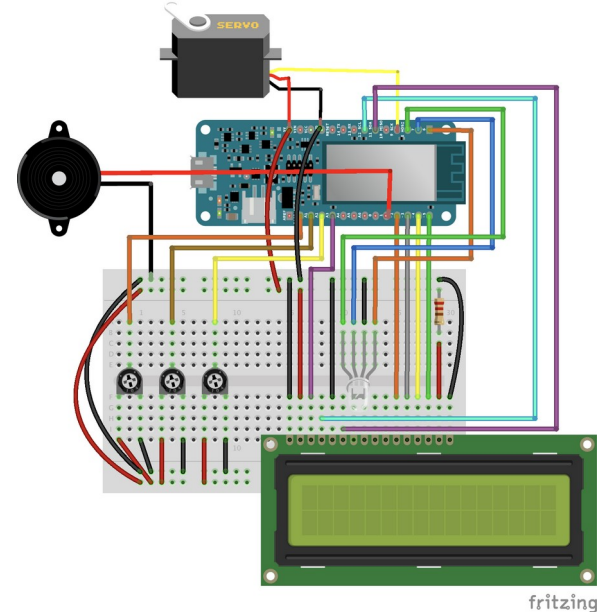


Sizing: $R = (V_{\text{supply}} - V_f) / I_{\text{target}}$

Different colors have different forward voltage V_f :

red ~1.8V, green ~2.1V, blue/white ~3.0V. Pick R per color; ~330 Ohm is a safe default.

In-class labs: controlled mechanism



Challenge: new system context

In-Class Structure

LECTURE

45–75 min

Conceptual overviews

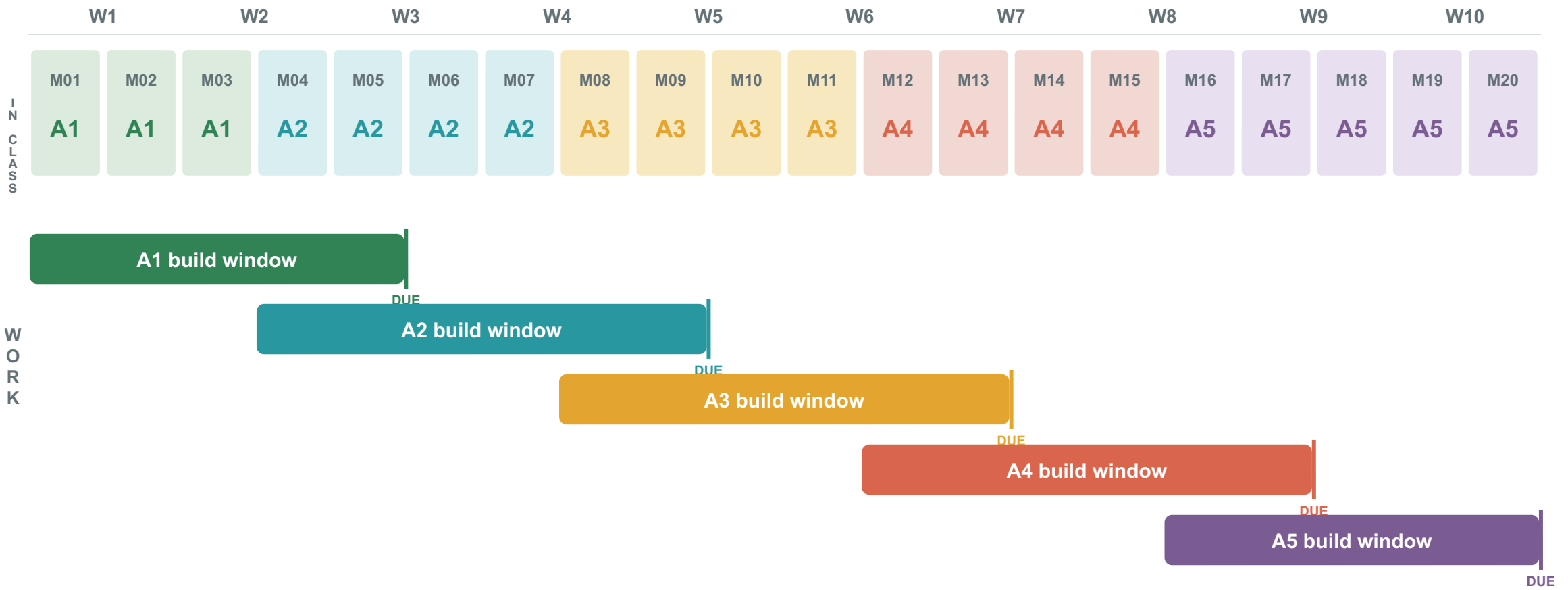
STRUCTURED LAB OR OPEN WORK

80–100 min

Bring the XIAO, laptop, a known-good data cable, and the components required for hardware work.

Course Cadence

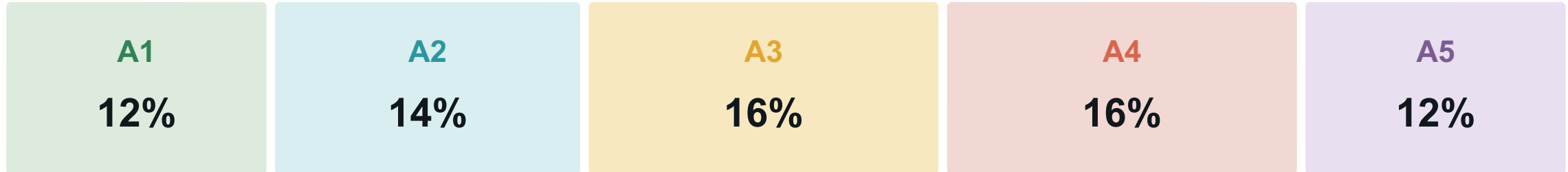
Concepts for the next assignment begin while the current assignment is still in progress.



Course Grade Components

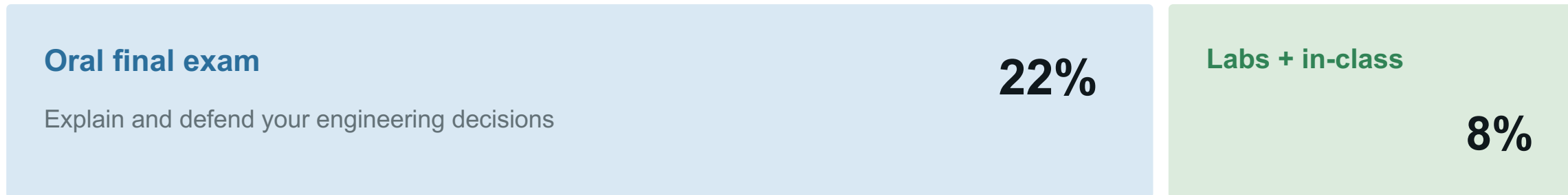
Assignments

70%



Individual explanation and participation

30%



Oral Final Exam



In-Person: 30-minute appointment during finals week

Course Resources

CANVAS
announcements +
logistics

SYLLABUS
policies + schedule

LECTURE + TEXT
transferable concepts

ASSIGNMENT HANDOUT
requirements + evidence

ECE 116 | FALL 2026

FALL 2026 | INSTRUCTOR WORKING DRAFT

ECE 116

Advanced Hardware and Software Design
Intelligent Embedded Systems

Instructor	Edward Wang ejaywang@ucsd.edu
Meetings	Tuesday and Thursday approximately three-hour blocks
Location	Official course schedule and Canvas
Course communication	Canvas announcements
Office hours and staff	Published on Canvas before the first meeting

Course description

ECE116 is a hands-on course about building intelligent embedded systems. We begin with a deterministic device that senses the physical world and responds over time. We then add a provided voice model, train a vision model, deploy that vision capability to a constrained microcontroller, and connect the device to VLM/LLM services through a secure edge-to-cloud architecture.

The course emphasizes engineering judgment. Students learn to distinguish a demonstration from evidence, measure behavior under documented conditions, identify failure at system boundaries, and defend a design under constraints such as latency, memory, energy, cost, reliability, security, and privacy.

No prior machine-learning experience is assumed. Machine learning is taught conceptually and through best practice. Mathematics is used when it helps make an engineering decision, not as a separate theory requirement.

Expected preparation

Students should arrive with introductory Python and basic Arduino experience associated with ECE18 or equivalent preparation. You should be comfortable with variables, conditionals, loops, functions, and running small programs. The course will teach the embedded architecture, evaluation, ML, TinyML, networking, and security concepts required for the assignments.

Formal prerequisite language will follow the approved UC San Diego catalog and enrollment rules.

Learning outcomes

By the end of the course, you will be able to:

ECE 116 | 1

Late Policy

FRI 7:00 PM

FRI 11:59 PM

MON 7:00 PM



posted deadline

automatic grace ends

submission closes

Full Credit

25% late deduction

No Credit

Collaboration Policy

WORK TOGETHER

discuss concepts
compare test strategies
interpret error messages
work near classmates

SUBMIT YOUR OWN

design + code
data + measurements
evidence + figures
written + oral reasoning

Do not exchange ready-to-submit work or use another student's measurements as your own. University academic integrity policies apply.

AI Use in This Course

Allowed

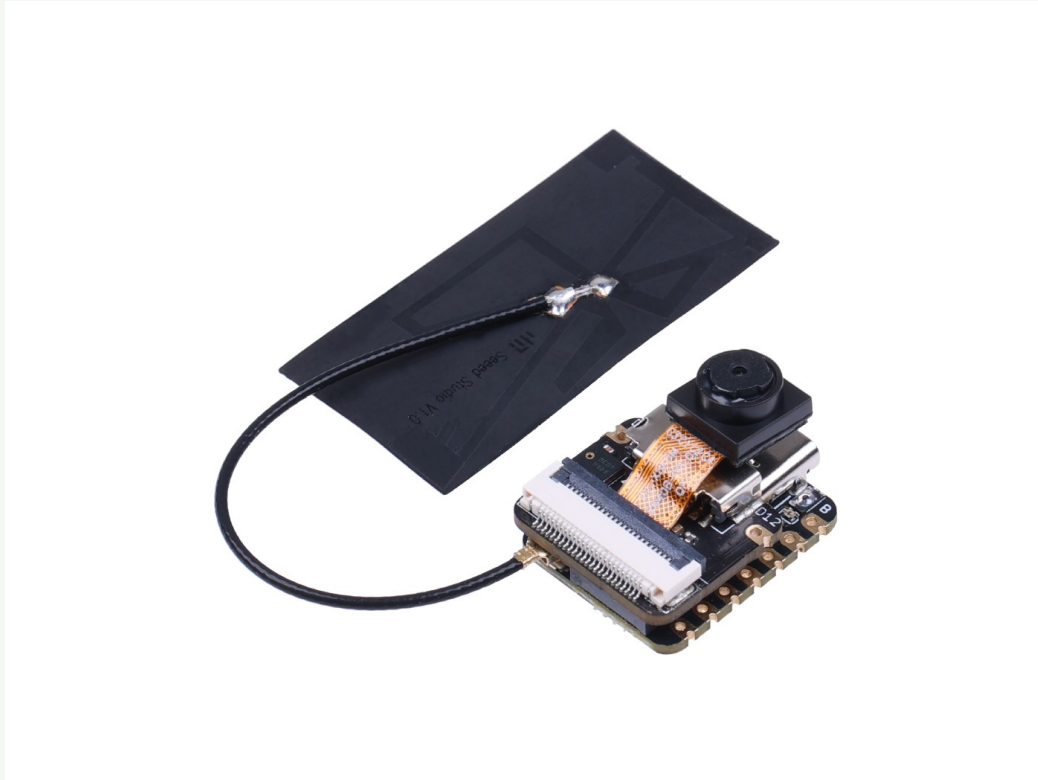
- ✓ Ask for an explanation of a concept
- ✓ Debug a specific error or compare approaches
- ✓ Improve clarity after you understand the work
- ✓ Use feedback to revise your own design

Not allowed

- ✗ Submit generated work you cannot explain
- ✗ Fabricate measurements, screenshots, or evidence
- ✗ Use AI during an oral assessment
- ✗ Bypass collaboration or citation requirements

You remain responsible for every design choice, result, and explanation you submit.

Lab 0: Get XIAO Working



compile

upload

serial

LED

Bring: XIAO + known-good USB data cable + current components

A known-good checkpoint turns later failures into diagnosable failures.